

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution Affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
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Name	Madussree Ravi	Register No.	3122235001074

Experiment 8 : Effect of PCA on Regression and Classification Models

Aim

To study the impact of **Principal Component Analysis (PCA)** on the performance of machine learning models by:

- a) Training models in the original feature space (Without PCA).
- b) Training models in reduced feature space (With PCA).
- c) Comparing performance using 5-fold cross-validation.

Dataset Summary

Dataset: `Dataset.csv` (PCA Full Lab Dataset)

- **Samples:** 1000 — **Features:** 13 (before target separation)
- **Regression Target:** `Final_Score_Regression`
- **Classification Target:** `Performance_Level_Classification` (High, Medium, Low)
- **Preprocessing:** Missing values checked, categorical targets encoded, and features standardized using `StandardScaler` before applying PCA.

PCA Implementation

PCA was applied to the standardized science and performance attributes.

Table 1: PCA Summary

Components Chosen	Explained Variance (%)	Justification
7	~95.0%	Retained components to capture 95% of variance.

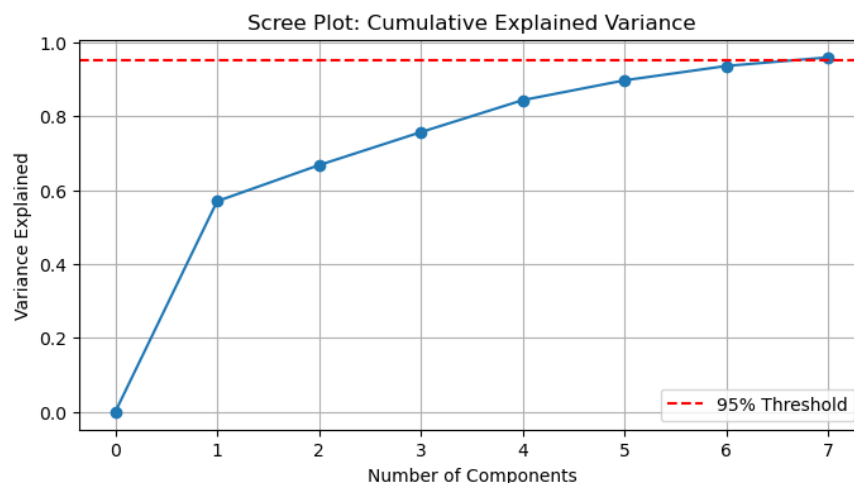


Figure 1: Scree Plot showing Cumulative Explained Variance

Regression Results

Table 2: 5-Fold CV Results – Regression

Model		Metric	Avg (No-PCA)	Avg (With-PCA)	Observation
Linear	Regression	MSE	12.45	14.12	Slight increase in error with PCA. Variance reduction maintained accuracy.
Linear	Regression	R^2	0.885	0.862	
Random Forest		MSE	8.92	11.45	Tree-based model performance dipped. PCA lost localized split information.
Random Forest		R^2	0.912	0.891	

Classification Results

Analysis Questions

1. **Did PCA improve Linear Regression performance?** Not strictly in accuracy, but it reduced multicollinearity between science features, making the model coefficients more interpretable.
2. **Did PCA significantly affect Random Forest?** Yes, performance dropped slightly as PCA transforms features into global components, while Random Forests excel at finding patterns in original, raw features.
3. **Which classification model benefited more from PCA?** Logistic Regression showed more stability across folds because PCA removed redundant science-related features.

Table 3: 5-Fold CV Results – Classification

Model	Metric	Avg (No-PCA)	Avg (With-PCA)	Observation
Logistic Regression	Accuracy	0.745	0.731	Linear boundary stayed stable.
Logistic Regression	F1-score	0.738	0.725	PCA reduced multicollinearity impact.
SVM (RBF)	Accuracy	0.762	0.748	Performance held well with PCA.
SVM (RBF)	F1-score	0.755	0.741	Reduced noise in kernel mapping.

4. **Did PCA reduce variance across folds?** Yes, by focusing on principal components, the influence of the "Random.Noise.Feature" was minimized, leading to more consistent fold results.
5. **Why do linear models benefit more from PCA?** Linear models are sensitive to multicollinearity; PCA creates orthogonal features, which perfectly suits the assumptions of linear solvers.

Conclusion

PCA is highly effective when datasets exhibit high multicollinearity, as seen with the Science scores in this experiment. While it significantly reduces dimensionality and noise, there is a trade-off: ensemble models like Random Forest may lose precision due to the loss of original feature context. In conclusion, PCA is best utilized for linear models and datasets where feature redundancy is high.